

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	06 July 2026
Team ID	
Project Name	Ledgerline – Credit Card Approval Prediction System
Maximum Marks	4 Marks

Technical Architecture: The deliverable includes the architectural diagram and the technology information per the tables below. The system follows a 3-tier architecture: Presentation (browser-based UI), Application (Flask + ML logic), and Data (CSV files + serialized model artifacts).

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	How the user interacts with the application — three browser-based pages: new application form, batch CSV upload, and model report	HTML, CSS (custom ledger design system), Jinja2 templates served by Flask
2	Application Logic-1	Web routing and request handling	Python, Flask
3	Application Logic-2	Feature engineering and preprocessing (converts raw form/CSV inputs into the model-ready feature vector)	pandas, NumPy, scikit-learn StandardScaler
4	Application Logic-3	Model training, comparison, calibration, and inference	scikit-learn (Logistic Regression, Decision Tree, Random Forest), XGBoost
5	Database	Applicant and credit history source data	CSV files (application_record.csv, credit_record.csv) — matches the Kaggle dataset schema
6	Cloud Database	Saved model artifacts for inference	Local filesystem (models/) for this build; portable to IBM Cloud Object Storage
7	File Storage	Saved model, scaler, feature schema, and chart files	Local filesystem (models/, static/)
8	External API	Optional cloud scoring endpoint for the trained model	IBM watsonx.ai (Watson Machine Learning) REST API
9	Machine Learning Model	Predicts credit card approval (Approve/Reject) from applicant features	Best of 4 trained classifiers, selected by ROC-AUC — Logistic Regression in this build
10	Infrastructure (Server / Cloud)	Local: Flask development server. Cloud: IBM Watson Machine	Local Flask server for development; IBM watsonx.ai for cloud deployment

		Learning online deployment endpoint	
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Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	All frameworks and libraries used are open-source	Flask, scikit-learn, XGBoost, pandas, NumPy, matplotlib, seaborn, Jupyter
2	Security Implementations	Server-side form validation; no API keys or model internals exposed via responses; no hard-coded credentials	Python input validation, server-side rendering (Jinja2)
3	Scalable Architecture	Stateless prediction logic (model loaded once at startup) allows horizontal scaling; cloud deployment via IBM Watson ML handles high request volumes	Flask (stateless), IBM watsonx.ai
4	Availability	Designed as a standalone service; containerizable for continuous availability; IBM Watson ML provides always-on cloud endpoint	Flask service, IBM Watson Machine Learning
5	Performance	Single-applicant predictions return in under 1 second; batch screening processes dozens of applicants in a few seconds	scikit-learn model inference (< 1ms per prediction)